

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

CHARUSAT Campus, Changa – 388 421.

M Sc (BIOTECHNOLOGY) SEMESTER II EXAMINATION BT710: Computational Skills

Date: 07.05.2011

Time: 10.00 AM to 11.30 AM

Total Marks: 35

Instructions to the candidates:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

PART A

Total marks: 10

Q1. Choose the correct answer from the options given below each question and put $\checkmark$ mark in front of it:

1. Third generation computers are
 - a) Vacuum tubes
 - b) Microprocessors
 - c) Integrated Circuits
 - d) Artificial Intelligence
2. PDA is a
 - a) A small handheld device
 - b) A Super computer
 - c) A mainframe
 - d) A Minicomputer
3. Which DOS command is used to create a directory?
 - a) mkdir
 - b) md
 - c) create dir
 - d) type dir
4. What is short cut (hot key) for slide show in MS power point?
 - a) F2
 - b) Ctrl + H
 - c) F5
 - d) F3
5. Each spreadsheet contains _____ rows.
 - a) 65,536
 - b) 256
 - c) 65,256
 - d) 1024
6. Which symbol is used for entity in ER model of database?
 - a) 
 - b) 
 - c) 
 - d) 
7. Primary Key _____
 - a) Uniquely identifies a record in table
 - b) Optimizes the table
 - c) Provides security to a table
 - d) Provides not null constraint for a table
8. Which C Function is used to the clear screen of console output.
 - a. clr();
 - c. getch();
 - b. clrscr();
 - d. printf(" ");
9. If $\cos \theta = \frac{3}{5}$ then $\sin \theta =$ _____
10. If $f(x) = x^2 + 2x + 1$ then $f(x-1) =$ _____

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Part B

Total Marks: 25

Section I

Q.1 Attempt any two

(05)

(i) If $f(x) = \log\left(\frac{1+x}{1-x}\right)$ then prove that $f\left(\frac{2x}{1+x^2}\right) = 2f(x)$.

(iii) Prove that: $\tan\left(\frac{\pi}{4} + \theta\right) - \tan\left(\frac{\pi}{4} - \theta\right) = 2 \tan 2\theta$.

(iii) Find AB where $A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 5 & 6 \\ 2 & 7 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 5 & 6 \\ 5 & 0 & 9 \\ 4 & 2 & 0 \end{bmatrix}$.

Q.2 Attempt any two:

(05)

1. What is Computer? List the major types of Computers.
2. List and explain major types of storage devices.
3. What is OS? List major services provided by OS.

Section II

Q.3 Attempt any one:

(05)

1. List and explain major Database models.
2. What is the difference between data and information? Define Database and Database management system with example.

Q.4 Attempt any one:

(05)

1. What is flow chart? Explain major symbols used in it with any example.
2. Write a program in C to print "Hello World". Explain its keywords, syntax and function used in your program.

Q.5 Attempt any one:

(05)

1. Which are application areas of Bioinformatics. Explain any two example of it related to Biotechnology.
2. What is Bioinformatics? Explain types of tools available for programming in it? Briefly explain any one example of tool.